

ĐẠI HỌC QUỐC GIA
THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC QUỐC TẾ

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập – Tự do – Hạnh phúc

Pre-thesis Registration Form

INSTRUCTIONS:

Student: Please fill your name and ID on the form and discuss with your Thesis Advisor to fill the other contents. Afterward, sign the form to verify your agreement and return the form to the Undergraduate Academic Assistant of the Department.

Thesis Supervisor: Please authorize the Thesis Registration Form for this student by signing below.

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Thesis title: Design and Implementation of a Digital Marketplace Platform for Engineering Services with Intelligent Multi-Criteria Service Matching.

Major: Network Engineering

Thesis goals and objectives:

This thesis aims to design and implement a digital marketplace platform that connects clients with engineering service providers through an intelligent multi-criteria service matching mechanism.

From a software engineering perspective, the system will be developed using structured requirement analysis, modular architecture design, and systematic testing methodologies.

From a network engineering perspective, the system will adopt a client-server architecture with secure communication protocols, optimized network interaction, and performance evaluation under different usage scenarios.

The objectives of this thesis include:

1. To analyze system requirements and design a scalable client-server architecture for a digital service marketplace.
2. To implement a modular and maintainable web-based system following software engineering principles.
3. To design and implement an intelligent multi-criteria service matching algorithm for ranking and recommending providers.
4. To implement secure authentication and role-based authorization mechanisms.
5. To design and evaluate network communication efficiency and system performance.
6. To conduct functional testing, performance testing, and basic security analysis.

Requirements:

(list all major requirements for the thesis work, refer to the guidance on the next page for details)

1. System Analysis and Architecture Design

- Requirement specification and system modeling (UML diagrams).
- Design of layered architecture (presentation, business logic, data layer).
- Client-server communication design.
- API design and documentation.

2. Database and Backend Development

- Relational database schema design.
- Implementation of data management and transaction handling.
- Data integrity and validation mechanisms.

3. Core Functional Modules

- User registration and authentication.
- Role-based Access Control (Client, Service Provider, Administrator).
- Service request submission and management.
- Service provider profile and skill management.
- Rating and feedback system.

4. Intelligent Multi-Criteria Service Matching

- Design of a weighted scoring model for provider ranking.
- Matching criteria may include:
 - Skill compatibility
 - Availability
 - Cost range
 - Location proximity
 - User rating
- Adjustable weighting mechanism.
- Comparison between basic matching and intelligent matching approaches.

5. Network and Security Considerations

- Secure client-server communication (e.g., HTTPS).
- Authentication and session management.
- Basic security mechanisms against common web vulnerabilities.
- Performance analysis of network requests (latency, response time).

6. System Evaluation

- Functional testing.
- Performance testing under different user loads.
- Evaluation of matching effectiveness.
- Discussion of system scalability and limitations.
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Name of Supervisor 1

Name of Student

Đinh Đức Anh Vũ

Phạm Vũ Quang

Date of Signed 8/02/2026

Date of Signed 8/02/2026

Name of Supervisor 2 (optional)

Name of Student 2 (optional)



Date of Signed

Võ Minh Thạnh

Date of Signed